FrameNet Semantic Role Classification by Analogy

Paper Review

Ngo, Afantenos, Lorini & Couceiro (arXiv 2603.19825, 2026)

Presented by DeMarko Fulcher | CMSC 612 | November 5, 2026

About Me



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Program: Computer Science M.S.

Research interests:

Formal Verification

Model Checking

Automated Reasoning

Schemas & Commonsense Reasoning

Schemas

- **Schema:** a mental template for familiar situations, with *slots* to fill
- Minsky's frames, Schank & Abelson's scripts
- FrameNet is a dictionary of schemas

Commonsense

- Understanding basic knowledge that relates to the context of a given situation
- They fill new slots by using **analogies** to situations we already know
- *Leaves feed trees* $\approx$ *lentils feed humans*: same schema

This paper: they fill FrameNet schemas by translating analogies to labelled examples

Key Concepts

Frame: a schema for a type of event or situation

Predicate : the word that *evokes* the frame

Frame Element : a slot, i.e. a participant

Semantic Role Classification : given the frame and the argument spans, label each span with its role

Hard Problem: ≈1,200 role types in FrameNet 1.7, where many have few examples to pull from

Frame: *Supply*

Lentils	**furnish**	proteins	to	humans
Supplier	predicate	Theme		Recipient

Two Views of Analogies

Proportional (standard NLP)

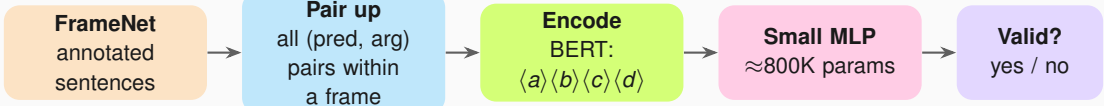
- *a* : *b* :: *c* : *d*, “a is to b as c is to d”
- *king : man :: queen : woman*
- Fixed word relations, **which can ignore context**
- Problem: *flower : petal :: tonne : kilogram* is considered “valid”

Relational (this paper)

- *(predicate, argument)* pairs from the **same frame**
- Valid $\Leftrightarrow$ arguments play the **same role**
- *provide : leaves :: furnish : lentils* ✓ (both Supplier)
- **Structure-Mapping Theory:** structure over surface

Sources: (Mikolov et al., 2013;)

Approach: Learning Analogies



Label: 1 if the arguments share a role, else 0

Method: The model is trained only to answer: "Do these two arguments play the same role?" then it compares the new argument with labeled examples and assigns the role that wins the majority vote.

Data: ≈6M balanced training pairs

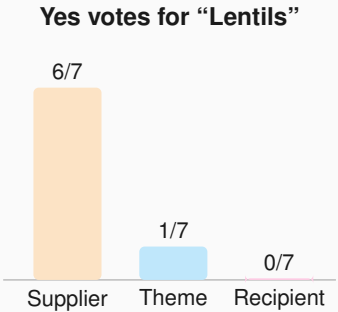
Cost: ≈7 min of training on a single V100 GPU

Inference: Analogical Transfer

New argument: **“Lentils”** (frame *Supply*)

1. Sample 7 labelled examples per role
2. Ask: *“Is this a valid analogy?”*
3. Count the yes votes per role
4. The role is decided based on a majority vote

A vote among examples, compared by *role*, not word overlap



Results on FrameNet 1.7

Setting	Prior	+ Analogy
Full pipeline	48.95	**49.81**
Open-SESAME, gold frames	58.46	**59.35**
Gold frames + spans, MLP baseline	20.64	**79.30**

Accuracy (%)

Consistent wins, but small: **+0.9 points** end-to-end

Binary model: 89.4% accuracy on test pairs

Unseen frames: only ≈4% drop

Unseen roles: ≈7% drop

Whole pipeline: ≈15 min, ≈50 Wh

Strengths

Open label set	Roles are never trained on, so a new role needs *examples*, not retraining which keeps flexible but impacts the system in other areas
Very Flexible	800K parameters, ≈50 Wh in total, versus fine-tuning BERT + GNNs pipelines
Explainable	Each decision can be traced back to the role that voted for it which can assist in better understanding of the results

Weaknesses

Small gains

+0.9 points, smaller than their own reproduction gap for (48.95 vs 47.85).

Weak baselines

The paper mentions feature-based system from 2002 (Gildea & Jurafsky) which reached about 80% on gold spans, so stronger reference points exist but its acknowledge that they use different versions of FrameNet and require more rigorous annotations.

Independent decisions

Each argument is labeled alone, so two spans can get the same core role and this is a weakness because in FrameNet each role should only be filled by one slot. It might be possible to enforce these constraints jointly with each other.

Sources: (Gildea & Jurafsky, 2002;)

Extensions: Other NLP Tasks

Relation Extraction
(head, tail) pairs in context. New relation = a few exemplars

Event Arguments
trigger : argument mirrors predicate : FE

Commonsense KBs
Classify event–inference links in ATOMIC by analogy

Narrative Schemas
Match event : protagonist pairs to predict the next event in a script

Cross-lingual SRL
English exemplars label low-resource languages via a multilingual encoder

Emerging Intents
Chatbots add a new intent by adding examples, with no retraining

Extensions: Improving the Method

Better Decoding

- Retrieve the most similar exemplars instead of random ones via a variance test
- Add joint constraints (one span per core role)
- Report variance over sampling seeds

Broader Comparisons

- Stronger encoders
- Compare to LLM in-context learning, which is also “reasoning by examples”
- Human study to test explainability

Key Takeaways

Idea

Schemas filled by **analogy**: “same role in the same frame”

Result

Tiny model, with small but consistent gains

Future work

Needs stronger baselines, variance tests, and joint constraints

Thank you!

Questions?

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